EXTENSION

Ag News

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Cost Effective Supplementation for Growing Calves

Mark Z. Johnson, Oklahoma State University Extension Beef Cattle Breeding Specialist

From a moisture standpoint, the summer of 2025 has been good for many Oklahoma cattle producers. As a result, many of us have ample standing forage.

This week we address the Oklahoma Gold and Oklahoma SuperGold supplementation programs as a means of adding profit potential. Both programs were designed to cost effectively improve the growth of calves and stocker cattle grazing pastures in late summer and fall. Both are based on limit feeding high protein supplements.

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Cost Effective Supplementation for Growing Calves
Mark Z. Johnson, Oklahoma State University Extension Beef Cattle Breeding Specialist (Continued)



Crude Protein (CP) content and digestibility of warm season grasses declines during mid-summer and fall. No doubt many Oklahoma producers watched green pastures turn brown during August when there was little rain coupled with several weeks of intense heat. This weather pattern corresponds with declining quality of standing forage. In late spring and early summer calves can gain 2 – 3 lbs./day on the same pastures that now may be limiting gains to less than half that amount, even with ample standing forage. Why? Because cattle's growth performance is based on the most limiting nutrient in their diet. The same grasses that would have contained over 10% CP in late spring and early summer are now likely well below that level. Regardless of our warm season forage type, (native grasses, Bermuda, sorghum Sudan or millet), the same maturity pattern holds true. A 500 lb. calf gaining 1.5 lbs./day requires a diet containing a minimum of 10% CP. Bottomline: at this point, CP has become the growth limiting factor in a calf's diet.

The Cost Effective Solution

If you have standing forage to be used before wheat pasture comes on, or just wanting to cost effectively add weight to growing calves, the key to improving growth rates is protein supplementation. As opposed to high energy creep feeds, Gold and SuperGold programs are designed to improve growth rate through high protein supplementation when ample standing forage is available. The Oklahoma Gold program includes a 38% CP supplement at a rate of 1 lb./day and includes an ionophore additive along with vitamins and minerals. The Oklahoma SuperGold program is similar as it includes supplementing a 25% CP at a rate of 2.5 lbs./day. The Oklahoma Gold program is one of the most cost effective and industry proven supplementation programs ever developed. From early June to October, several trials conducted at OSU, indicate feeding 1 lb./day of Gold can increase gains of stockers grazing summer pastures by up to 0.6 lb./day. Furthermore, the Gold program is based on a low volume of feed and offers the flexibility of feeding every-other-day. In some situations the SuperGold feeding program may offer advantages. Remember, both of these feeding strategies work based on ample standing forage. More details for both feeding strategies can be found at http://osufacts.okstate.edu in the fact sheets referenced below.

References:

Oklahoma Cooperative Extension Fact Sheet ANSI-3032. Oklahoma Gold Q&A.
Oklahoma Cooperative Extension Fact Sheet ANSI-3033. Oklahoma SuperGold Q&A.



WHEAT AND CANOLA NITROGEN MANAGEMENT

Josh Bushong, Area Extension Agronomist

Low grain prices should force farmers to be extra thoughtful on managing fertility plans for this year's winter crops. I've received a few questions about how much nitrogen should be applied and how much can be applied in-furrow. Forage and grain yield goals are always driven by nitrogen, but contingent on growing conditions. Luckily, we have some moisture in the soil profile to favor good wheat pasture.

Nitrogen (N) is a mobile nutrient and can be lost or become unavailable to plants. Source or type of N fertilizer used can also have a big impact. Different production systems will require more or less nitrogen up front than others. Optimal nitrogen application rates to increase the use efficiency will also vary from year to year and farm to farm.

In a dual-purpose or grazeout wheat system, more N is needed early compared to a grain only system to produce more forage. A grain only system needs about 2 lb N per bu of seed produced, or 80 lb N for a 40 bu grain yield. It takes about 60 lb of N to produce one ton of wheat forage. So in a dual-purpose system, 60-70 lb of N will be needed at planting compared to 30-40 lb of N for a grain only system. The second application of N is typically applied late fall to early spring.

Many grain drills and especially air seeders can apply fertilizer in-furrow directly with the seed wheat. Farmers with acidic, low pH, soils often use this practice to apply phosphorus to mitigate some availability issues and aluminum toxicity. To prevent germination issues, OSU recommends to keep the salt load (N and potassium) to no more than 30 pounds when sown on 6-8 inch ow spacing in medium to fine textured soils and 21 pounds on sandy or dry soils. Applying 167 pounds of 18-46-0 would yield 30 pounds of N.

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WHEAT AND CANOLA NITROGEN MANAGEMENT

Josh Bushong, Area Extension Agronomist

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It is not recommended to use fertilizer sources that are high in N and place it directly with the seed as it could cause more seed and seedling injury. When urea (46-0-0) is placed in-furrow it initially converts to ammonia. Ammonia can be toxic to plant roots and reduce seed germination. The risk of injury is higher in drier soils, sandier soils, and in higher pH soils. Field research from KSU has shown stand reductions of 32% when 30 pounds of N from urea was applied in-furrow. This resulted in a 14% grain yield reduction.

For grain only producers, the old rule of thumb of 2 lb N per bushel is a good starting place but that might be an overestimation. When looking at more than 15 years of field trial data from the OSU North Central Research Station near Lahoma, the optimum pounds of N per bushel ranged from 0 to 3.2. The average pounds of N per bushel to reach an economic optimum N rate was closer to 1.6, however if 2 pounds per bushel was applied the grain yield would have been maximized 13 out of those 15 years.

Split application is often a best management practice in canola as well. The canola crop needs enough N to produce a healthy crop to better withstand the winter. Too much N in the fall or at planting can result in excessively large plants going into winter. Excessive N and prolonged warm growing conditions can lead to winter survival issues if the plants growing point starts to vertically elongate. Canola needs about 2.5 lb N per bushel of seed yield, or 100 lb N for a 40 bu yield.

It is almost impossible to determine the total N needs at the time of planting to maximize grain yields. Topdressing N on wheat and canola is a good management practice because it decreases the risk of N losses as well as benefiting from influencing late season N recommendations based on the potential yield of the crop. Topdress application rates can be impacted by current expectations of the crop and weather forecasts. Basically, estimating the yield potential becomes more accurate as the season progresses.
Using tools at hand can dramatically influence N recommendations. Applying N-Rich strips to wheat in early fall can help estimate N demands throughout the season for both dual-purpose and grain-only systems. This management tool can assist in determining N deficiencies or sufficiency.

The N-rich strips can be as simple as hand spreading a few cups of urea or more technical with using custom built applicators on UTV's or tractors. The strips can simply be used to visually determine if there is enough N or not. If the strip cannot be seen, then there is a low likelihood of getting a response to additional N. If the strip can easily be seen, then more N is likely needed.



BLAINE COUNTY HOSTS A SUCCESSFUL FALL FREE FAIR

Blaine County celebrated a fantastic Fall Free Fair this August! Inside exhibits were judged on Friday, August 22, alongside the livestock judging contest. The excitement continued Saturday, August 23, with the livestock show and the always-popular tractor driving contest.

We extend our heartfelt appreciation to everyone who participated by entering exhibits and attending the fair. A special thank you goes out to our dedicated superintendents for their outstanding work.

The Blaine County Fairboard, in collaboration with the county's Ag Teachers, worked tirelessly to make this event a success. If you see a Fairboard member or an Ag Teacher, be sure to thank them for their hard work and commitment to our community.

We're also grateful to our judges—Sue Schmidt, Jean Bailey, Karen Armbruster, Sally Nighswonger, Jessica Nickels, Rick Nelson, and Carter Burgin—for lending their expertise. And a big thank you to Cassie Ayers for generously donating her time and capturing memorable moments throughout the fair.

Finally, we'd like to extend our sincere thanks to the Lions Club for running the concession stand during the fair. Your support and service helped make the event even more enjoyable for everyone.

Fair results can be found in your local newspaper, or by contacting the Blaine County OSU Extension Office.

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Dates to Remember



September 11-21
Oklahoma State Fair

September 25-October 5
Tulsa State Fair

October 10-11
Watonga Cheese Festival

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